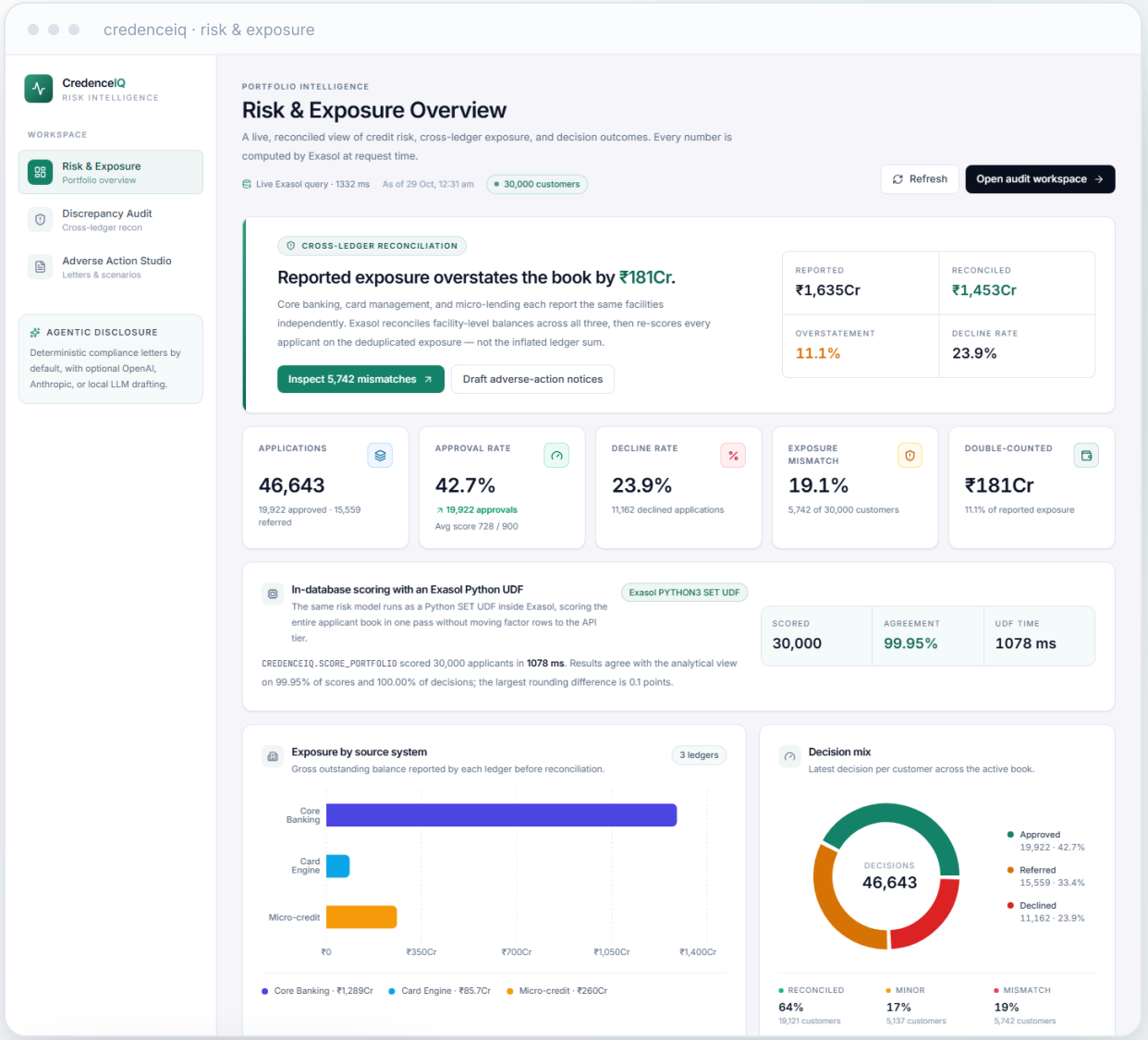
CredenceIQ
RISK INTELLIGENCE

EXASOL HACKATHON · 2026

Credit decisions you can defend.

CredenceIQ reconciles conflicting credit exposure, re-scores applicants inside Exasol, and turns declines into regulator-ready disclosure drafts — with a full audit trail from portfolio number to factor-level reason.

- Exasol Personal 2026.2
- Python SET UDF
- FastAPI
- Next.js 15



₹181Cr
DOUBLE-COUNTED EXPOSURE
11.1% of the reported book, surfaced by facility-level reconciliation.

`github.com/KrithickGanesh/credence-iq` · Team CredenceIQ

THE PROBLEM

The book doesn’t agree with itself.

01 **Three ledgers, one borrower**

Core banking, card management, and micro-credit each report the same facility independently — often with different outstanding balances.

02 **Exposure is overstated**

Double counting inflates the book, distorts utilisation and DTI, and pushes good applicants below the approval line.

03 **Declines arrive without a defence**

A score is not a reason. Lenders need principal factors, appeal rights, and a grievance route the customer can act on.

Seeded demo book: 30,000 customers · 115,733 exposures · 46,643 applications.



₹181.1Cr

EXPOSURE REMOVED BY RECONCILIATION

₹1,635Cr reported → ₹1,453Cr reconciled · 11.1% of the book.

19.1%

CUSTOMERS WITH A CROSS-LEDGER MISMATCH

5,742 of 30,000 customers. A further 5,137 have minor variance.

7,080

TRUE DECLINES IN THE CURRENT BOOK

Each one needs a defensible, auditable adverse-action notice.

THE SOLUTION

One reconciled credit 360 — inside the database.

Every analytical step runs in Exasol. The API shapes responses; it does not recompute the book in Python.

STEP 01

Reconcile

`V_RECONCILED_CREDIT_360` deduplicates facilities across core banking, cards, and micro-credit, then derives variance, utilisation percentiles, and a transparent score.

STEP 02

Re-score

One weight table drives three execution paths: the Python API, the SQL view, and a generated `PYTHON3 SET UDF` that scores all 30,000 applicants in-database.

STEP 03

Respond

Generate a formal adverse-action notice and a plain-English companion with factor disclosures, appeal rights, and the exact path back to approval.

Live SQL **No mocked core data.** Portfolio totals, applicant 360s, simulations, and notices are all driven by queries against the Exasol book.

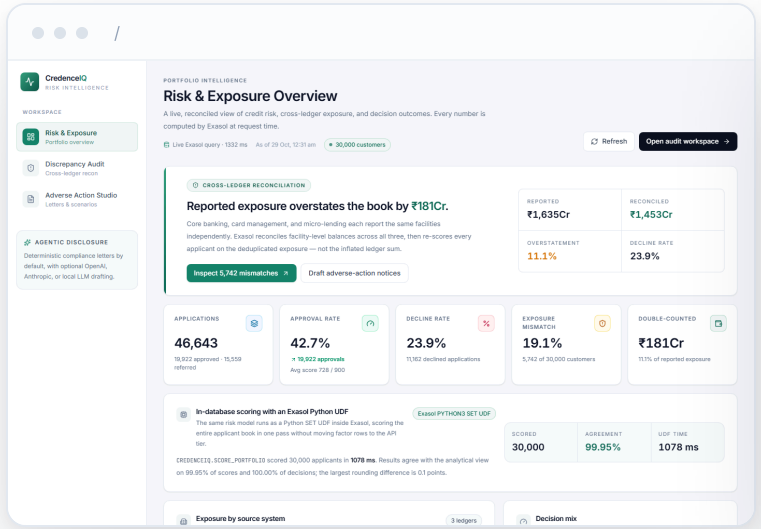
REPORTED BOOK
₹1,635Cr → ₹1,453Cr reconciled

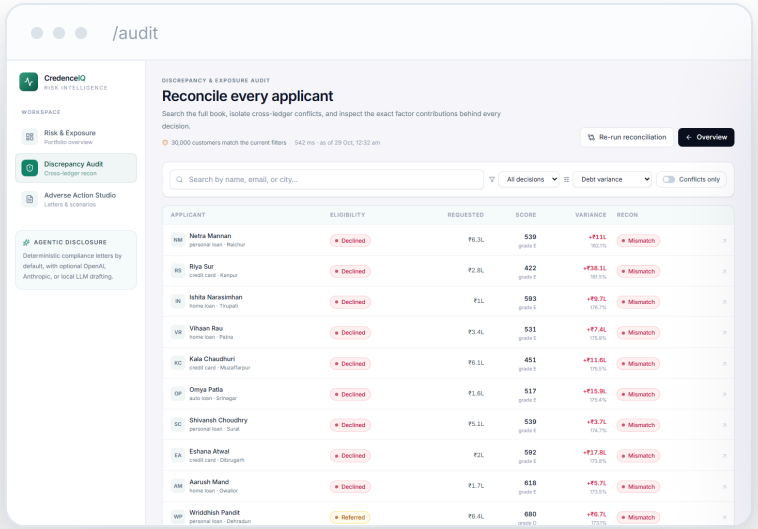
HISTORICAL DECISION
Re-scored on deduplicated balances

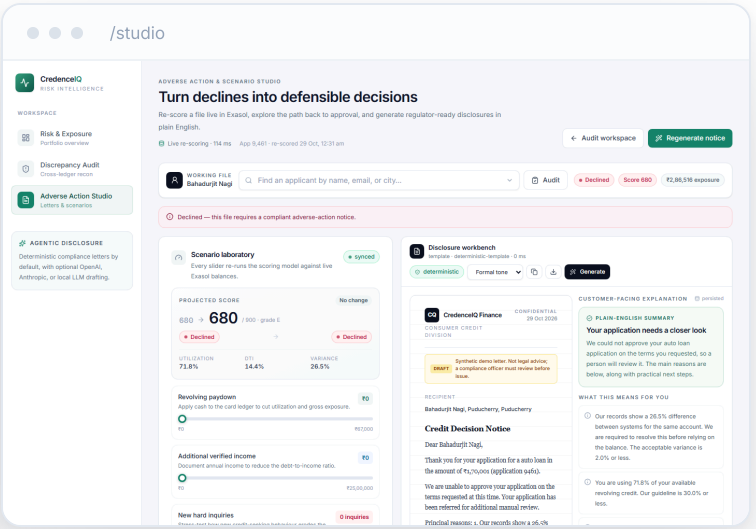
OPAQUE DECLINE
Principal reasons + appeal route

THE PRODUCT

A workspace for risk, audit, and compliance.







01 · MONITOR

Risk & exposure

Reconciled KPIs, decision mix, rejection velocity, denial drivers, and the largest unresolved exposures.

02 · INVESTIGATE

Discrepancy audit

Search the book, isolate conflicts, compare historical vs. re-scored decisions, and open the Applicant 360 drawer.

03 · ACT

Adverse-action studio

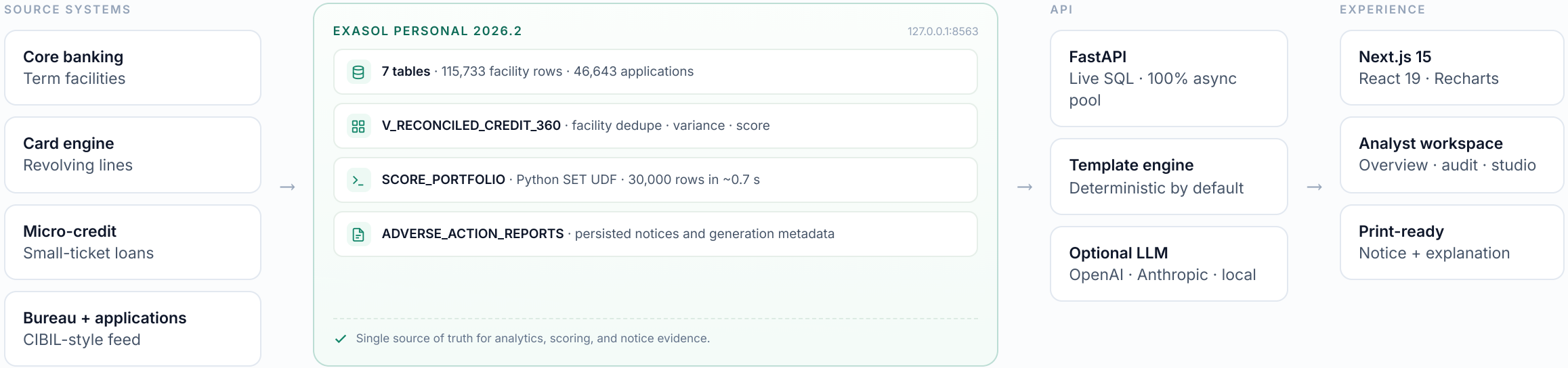
Re-score a file live, explore the path to approval, and generate the formal notice plus plain-English explanation.

Applicant 360

Portfolio → applicant → facility → factor. Every figure in the UI is traceable to the query and the record behind it.

ARCHITECTURE

Built around Exasol, not on top of it.



`Python 3.12` `pyexasol 3` `Exasol SLC python-3.12` `FastAPI` `Pydantic 2` `Next.js 15.5` `TypeScript` `Tailwind 3.4` `Docker Compose`

PERFORMANCE

400 round-trips → one set-based pass.

99.0%

LOWER WALL TIME · 5,190.9 MS SAVED

The same 100-customer reconciled result, composed two ways over the same Exasol data. One set-based credit-360 query replaces four hundred sequential round-trips.

5,244.7 ms — N+1 COMPOSITION · 400 ROUND-TRIPS

53.8 ms — SINGLE PASS · 1 ROUND-TRIP

PATH	ROUND-TRIPS	MEDIAN	MS / CUSTOMER
Per-customer N+1	400	5,244.7 ms	52.4
Single set-based pass	**1**	**53.8 ms**	**0.5**
Portfolio aggregate scan	1	131.7 ms	—

Honest benchmark. This measures query composition over the same Exasol data — not Exasol against another engine. Warm connections, 100-customer sample, median of 3 runs.

30,000	99.95%	100%	131.7 ms
APPLICANTS SCORED IN-DATABASE	SCORE AGREEMENT WITH THE VIEW	DECISION AGREEMENT	PORTFOLIO AGGREGATE SCAN

COMPLIANCE BY DESIGN

Explainability is a regulatory requirement.

Every decline carries

- Principal reason codes with factor weights
- The observed value, benchmark, and score impact
- Bureau, utilisation, DTI, and exposure disclosures
- Appeal rights and a named grievance route
- A plain-English companion with next steps
- Provider, model, and generation mode persisted

Draft, not legal advice. Deterministic by default; optional hosted or local LLM. Every notice says so.

Citations verified against official sources

RBI Credit Information Reporting Directions
2025 · bureau use, dispute, correction

RBI Digital Lending Directions
2025 · disclosures, borrower communication

Fair Practices Code
reasonable, non-discriminatory reasons

Integrated Ombudsman Scheme
2021 · escalation route

CICRA
2005 · credit-information rights

DPDP Act + Rules
2023 / 2025 · purpose, notice, grievance

Synthetic demo data | No production PII | Model weights auditable

Anatomy of a generated notice

Persisted with provider, model, mode, and factor payload

01	02	03	04	05
Principal reason	**Evidence**	**Impact**	**Appeal**	**Grievance**
Factor code and weight	Observed value vs benchmark	Points contributed	Rights and review route	Named contact and escalation

INDIA LOCALISATION

Built for the Indian lending stack.

CURRENCY

₹1,88,643 · ₹1,453Cr

Indian digit grouping throughout — lakh and crore on dense screens, full rupees in notices.

BUREAU SCALE

300–900 · approve 750 · refer 680

CIBIL-style model score, directly comparable to the bureau benchmark of 680.

IDENTITY

en_IN · curated city/state map

Names, cities, and addresses are internally consistent — no "Gudivada, Madhya Pradesh".

DISCLOSURE

RBI + DPDP · factor metadata

Reason codes cite the instrument behind them; notices carry appeal and grievance routes.

PLAIN-ENGLISH COMPANION · LIVE DRAFT

"Our records show a 9.4% difference between systems for the same account. We are required to resolve this before relying on the balance. The acceptable variance is 2.0% or less."

Generated from the factor payload · deterministic template

DEMO PATH · CUSTOMER 129798

675.5 → 778.9

Declined → Approved after the full ₹22.7K card payoff, mismatch resolution, and ₹2.5L verified income. Each lever alone stays below 750.

SCORE JOURNEY · CUSTOMER 129798

+103.4 points



WHY IT MATTERS

Less reconciliation. More defensible decisions.

Risk teams

One reconciled view of the book. Re-score the entire portfolio in the database instead of stitching four queries per customer.

Compliance

Consistent reason codes, factor evidence, appeal rights, and a persisted audit trail for every notice generated.

Borrowers

Clear reasons, the figures behind them, and a documented path back to approval — including the exact paydown gap.

5,742	30,000	99.0%	100%
MISMATCHES SURFACED	APPLICANTS SCORED IN-DATABASE	LOWER WALL TIME	UDF DECISION AGREEMENT

Next: bureau and Account Aggregator connectors · model governance and drift monitoring · multi-tenant lender deployment · configurable notice templates.

Three ledger views → **facility-level truth**

Four queries per customer → **one set-based pass**

Ad-hoc decline emails → **persisted, factor-level notices**

Success metrics: reconciliation coverage · mismatch resolution time · notice generation latency · appeal and override rate.

RUN IT LOCALLY

The whole book is reproducible.

Exasol Personal, a deterministic seed, and two app processes. The README has the full quickstart and the fallback path.

powershell · quickstart

```
# 1 · database
exakit start
exakit status

# 2 · deterministic book
python seed_data.py --if-empty --customers 30000

# 3 · API + dashboard
uvicorn app.main:app --port 8000
npm run dev
```

REPOSITORY
github.com/KrithickGanesh/credence-iq

Source, deck, and demo
Scan to open the public repository and the full run instructions.

Source code	FastAPI · Next.js · seed
README	quickstart + architecture
Pitch deck	docs/pitch-deck.pdf
Demo video	docs/demo/credenceiq-demo.mp4
Benchmark	scripts/benchmark.py

CredenceIQ

Synthetic demonstration data · not legal advice · MIT licensed